LAMP-Merge: Long-tail-Aware Medical Prototype Merging for One-shot Asynchronous Clinical Models

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Abstract

Model merging aims to integrate multiple independently trained local checkpoints into a unified global model, typically through general-purpose paradigms such as parameter interpolation, task-vector composition, or update-direction alignment. These methods implicitly assume that client models share relatively complete and comparable global discriminative structures. We argue that this assumption does not hold in medical settings, giving rise to two diagnostic-structure anomalies: (1) **Aggregation-induced prediction collapse after merging**: When clients have incomplete diagnostic-class coverage, multiple locally effective models can be merged into a globally collapsed model whose predictive distribution concentrates on a small number of dominant classes. (2) **Long-tailed medical prevalence cannot be ignored**. Majority classes in medical data often reflect genuine clinical prevalence rather than merely data noise. To address these challenges, we propose **LAMP-Merge**, which recasts medical model merging from global parameter interpolation as class-level diagnostic-evidence reconstruction. Through **diagnostic prototype reconstruction(DPR)** and **long-tail prevalence calibration(LPC)**, LAMP-Merge constructs a global diagnostic model without exposing raw medical images or requiring multi-round federated communication. Experiments on the Blood, Derma, Organ-C, Organ-S, and Ultrasound medical imaging datasets show that LAMP-Merge outperforms existing model-merging methods in most client-averaged settings while substantially mitigating prediction collapse.

Introduction

Federated learning (McMahan et al. 2017; Li et al. 2020; Karimireddy et al. 2020; Kairouz et al. 2021) and model merging (Izmailov et al. 2018; Wortsman et al. 2022; Ilharco et al. 2023; Yadav et al. 2023) provide two important collaboration routes for privacy-sensitive multicenter medical intelligence. Federated learning enables hospitals to train collaboratively without sharing raw data (Sheller et al. 2020; Rieke et al. 2020), while model merging further reduces communication and optimization costs by integrating independently trained local models after training (Ainsworth, Hayase, and Srinivasa 2023; Stoica et al. 2024; Wu et al. 2025). These paradigms are particularly attractive for **medical imaging**, because a single hospital rarely covers the full disease spectrum, while privacy regulations and device heterogeneity make centralizing raw medical images difficult (Kaissis et al. 2020a; Bonawitz et al. 2019; Abadi et al. 2016).

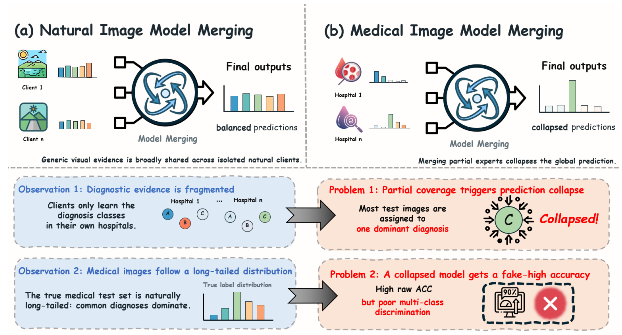


Figure 1: **Motivation.** Unlike natural-image clients that share overlapping visual evidence, medical clients each observe only part of the diagnostic space, so parameter-level merging collapses the global model onto a few dominant classes. This reflects two observations: diagnostic evidence is fragmented across isolated clients, and medical prevalence is long-tailed—under which a collapsed model can still report deceptively high accuracy.

Despite their success on natural images and general visual tasks (Izmailov et al. 2018; Wortsman et al. 2022; Ilharco et al. 2023; Entezari et al. 2022; Ainsworth, Hayase, and Srinivasa 2023), transferring these paradigms to medical data remains challenging. A key reason is that general-purpose model merging implicitly assumes that different client checkpoints contain relatively complete and comparable **global discriminative structures**; local models are expected to align through parameter space, task-vector space, update signs, or curvature statistics (Yadav et al. 2023; Yu et al. 2024; Wu et al. 2025; Jin et al. 2025). This assumption does not hold in medical scenarios. Medical centers differ not only in device domains but also in diagnostic-class availability: some centers may mainly cover common diseases or specific examination types, while having few rare-disease samples (Guan and Liu 2022; Zech et al. 2018; Li et al. 2021). The resulting local models are not complete replicas of the global task, but **partial diagnostic experts** supported by locally observed diseases.

As illustrated in Fig. 1, we summarize these challenges as two diagnostic-structure abnormalities in medical model merging:

- **Diagnostic evidence fragmented across isolated medical clients.** Each client has reliable supervision only for observed classes; thus, a local model may be locally discriminative but still lack evidence for the full global diagnostic space (Guan and Liu 2022; Li et al. 2021).
- **Long-tailed Medical Image Prevalence.** Medical image datasets naturally follow a long-tailed distribution: a few common diagnostic classes dominate the samples, while rare classes are underrepresented (Buda, Maki, and Mazurowski 2018; Brodersen et al. 2010). Under this imbalanced class prior, a merged model collapsed to the majority class can still obtain high Accuracy despite lacking complete multi-class discrimination.

To address these issues, we propose **LAMP-Merge, a long-tail-aware medical prototype-merging method for single-shot asynchronous medical image model merging**. LAMP-Merge shifts the merge target from global parameter interpolation to **class-level diagnostic-evidence reconstruction**. The key idea is to use client-uploaded class-level statistics as a synchronization signal for the global diagnostic space, allowing the server to reconstruct a global diagnostic classifier without accessing raw medical images, using a server-side validation set, or conducting multi-round federated communication. Specifically, LAMP-Merge consists of two mechanisms: **diagnostic prototype reconstruction** aggregates reliable class-level evidence through prototypes in a shared reference feature space, and **long-tail prevalence calibration** injects a bounded prior bias from client-uploaded class counts, thereby mitigating prediction collapse while preserving clinically meaningful majority-class prevalence.

Our main contributions are summarized as follows:

- **Medical Merging Diagnosis.** We revisit model merging under single-shot asynchronous medical collaboration and identify limited local diagnostic coverage and aggregation-induced prediction collapse as two key abnormalities caused by medical class availability and long-tailed prevalence.
- **LAMP-Merge.** We propose a long-tail-aware medical prototype-merging method that reconstructs a global diagnostic classifier from client-side class prototypes and prevalence statistics without exposing raw images or requiring multi-round federated optimization.
- **Empirical Validation.** We conduct full-scope experiments on five medical image datasets, four vision-model families, three client-population sizes, and three non-IID skew levels, showing that LAMP-Merge consistently outperforms strong model-merging baselines and improves prediction-collapse diagnostics.

Related Works

Model Merging

Model-merging methods integrate independently trained checkpoints in parameter, task-vector, or geometric spaces. Weight averaging (Izmailov et al. 2018) and Model Soups (Wortsman et al. 2022) directly interpolate parameters; permutation-aware matching (Entezari et al. 2022; Ainsworth, Hayase, and Srinivasa 2023) aligns hidden-unit symmetries; ZipIt (Stoica et al. 2024) performs feature-level zipping; Task Arithmetic (Ilharco et al. 2023), TIES-Merging (Yadav et al. 2023), and DARE (Yu et al. 2024) manipulate task deltas and sign conflicts; Fisher merging (Wu et al. 2025), RegMean (Jin et al. 2025), Model Stock (Jang, Yun, and Han 2025), Breadcrumbs (Davari and Belilovsky 2024), Iso-C (Marczak et al. 2025), FreeMerge (Zheng and Wang 2025), and RobustMerge (Zeng et al. 2025) introduce curvature, mean-statistic, subspace, frequency-domain, or robustness assumptions.

However, these methods share a class-agnostic merging interface: their variables are global parameters, task vectors, update signs, or geometric statistics rather than class-level diagnostic evidence. **They do not encode whether a client has reliable supervision for a specific medical class.** Under incomplete local class coverage and long-tailed prevalence, reliable class directions can be mixed with absent or conflicting directions from other clients, resulting in aggregation-induced prediction collapse(Buda, Maki, and Mazurowski 2018; Zech et al. 2018).

One-shot clinical merging therefore requires class availability to be distinguished from model-wide parameter similarity.

Medical Model Merging

Medical fusion studies exploit domain structures such as irregular clinical time series, physiological sensor graphs, and image-EHR fusion. mTAN (Shukla and Marlin 2021), Raindrop (Zhang et al. 2022), and MedFuse (Hayat, Geras, and Shamout 2023) show that medical tasks benefit from modeling clinical sampling mechanisms, modality heterogeneity, and patient-level temporal relations.

However, their fusion objects are usually patient-level inputs, multimodal records, or clinical representations rather than independently trained image-classifier checkpoints. They often require patient samples, timestamps, electronic health records, or server-side validation feedback, which conflicts with common privacy constraints (Kaissis et al. 2020a; Sheller et al. 2020; Bonawitz et al. 2019). MedMerge (Wei et al. 2026) is a relevant training-free medical knowledge-transfer framework, but it targets cross-modal adaptation for medical vision-language models rather than asynchronous checkpoint merging for the same classification task. **Thus, existing medical fusion routes are mismatched with our privacy and single-upload protocol constraints.** LAMP-Merge keeps the medical structure uploadable by using only locally computed class prototypes and prevalence counts.

Federated Learning

Federated and decentralized learning perform collaborative training through communication protocols. FedAvg (McMahan et al. 2017) and its variants or medical deployments (Li et al. 2020; Karimireddy et al. 2020; Li et al. 2021; Kaissis et al. 2020b; Sheller et al. 2020; Rieke et al. 2020; Kairouz et al. 2021) rely on repeated server-client updates; Swarm

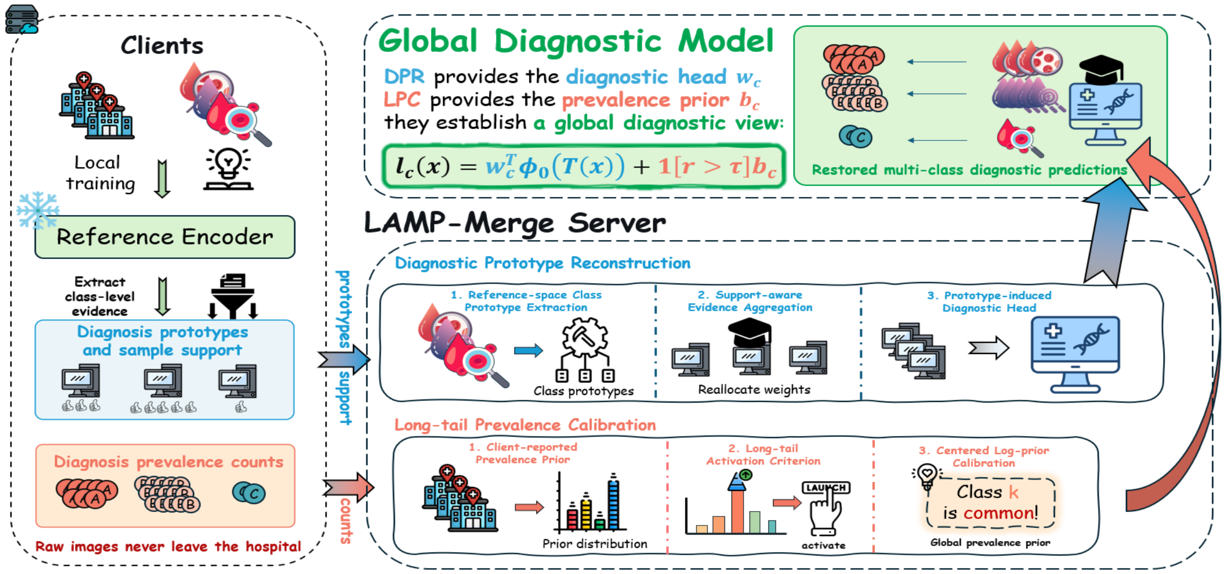


Figure 2: **Overview of LAMP-Merge.** Each hospital uploads only class-level statistics—reference-space prototypes with sample support and diagnosis prevalence counts—while raw images stay local. On the server, Diagnostic Prototype Reconstruction (DPR) aggregates the prototypes by support-aware weighting into a diagnostic head $\mathbf{w}_c$, and Long-tail Prevalence Calibration (LPC) adds a gated, bounded log-prior bias b_c, giving the scoring rule $\ell_c(x) = \mathbf{w}_c^\top \phi_0(T(x)) + \mathbf{1}[r > \tau]\, b_c$ in a single-shot merge without raw-data sharing or federated rounds.

Learning (Warnat-Herresthal et al. 2021) and Gossip Learning (Heged\Hus, Danner, and Jelasity 2019) reduce centralization but still require participants to continue training, exchanging models, and optimizing from intermediate states.

However, our setting begins after local training has finished. Hospitals may not remain online, receive global checkpoints repeatedly, or run additional optimization rounds. **The bottleneck is therefore communication feasibility in addition to privacy.** LAMP-Merge requires one upload of a local model interface and class-level statistics, enabling the server to construct a unified diagnostic model without a multi-round feedback loop.

Method

In this section, we first formulate the diagnostic-structure pathologies exposed when generic model merging is applied to medical data, namely **limited local diagnostic coverage** and **aggregation-induced prediction collapse**. To correct these intrinsic mismatches, we propose **LAMP-Merge**, a two-stage framework composed of **DPR** and **LPC**, as shown in Fig. 2.

Preliminaries

Problem Formulation. Assume K medical clients. Client i owns a private labeled dataset $D_i = \{(x, y)\}$ drawn from a class-skewed medical distribution over C diagnosis classes. Let $D_{i,c}$ denote the subset of client i belonging to diagnosis class c. In medical partitions, $D_{i,c}$ can be empty for many classes; therefore, client i has reliable supervision only for its observed classes. The goal is to construct a global diagnostic model under this partial and heterogeneous diagnostic-label coverage.

Upload and Server Constraints. After local training, each client uploads its checkpoint interface and class-level statistics once; the server receives no raw images, per-sample data, validation set, or iterative feedback. The uploaded support $n_{i,c}$, prevalence $m_{i,c}$, and reference-space prototype $\mu_{i,c}$ enable one merge with class-level diagnostic evidence as an explicit variable.

Motivation

Although generic model merging is training-free and parameter-efficient, we find that its diagnostic structure can severely degenerate in medical model merging. Prediction-distribution diagnosis and class-support analysis reveal two key abnormalities:

(1) Aggregation-induced prediction collapse. Under partial diagnostic coverage, parameter-level merging can turn several locally useful predictors into **a globally collapsed predictor**. The merged model tends to allocate most test images to a few dominant classes, indicating that global class-discriminative structure has not been reconstructed.

(2) Prevalence prior cannot be ignored. Medical datasets often exhibit clinically meaningful long-tailed prevalence. Therefore, completely suppressing majority-class dominance is also inappropriate: a collapsed predictor is undesirable, but a merger should still preserve calibrated disease prevalence when the dominant class reflects the true clinical prior.

Diagnostic Prototype Reconstruction Module

To address **fragmented diagnostic evidence across isolated medical clients**, DPR reconstructs class-conditional evidence in a shared reference space and synthesizes a global diagnostic head over the full label space.

(1) Reference-space Class Prototype Extraction. First, all clients extract class prototypes with the same frozen reference backbone. Let ϕ_0 be the reference backbone instantiated from the shared model configuration before local training, and let $T(\cdot)$ be the evaluation transform. For each diagnosis class c, client i locally computes a reference-space class prototype

$$\mu_{i,c} = \frac{1}{n_{i,c}} \sum_{(x,y)\in D_i} \mathbf{1}[y=c]\phi_0(T(x)), \tag{1}$$

where $n_{i,c} = |D_{i,c}|$ is the class support count. If $n_{i,c} = 0$, the client uploads zero support for that class and contributes no prototype evidence.

(2) Support-aware Evidence Aggregation. Second, the server converts class-support counts into evidence weights to characterize the reliability of each client for a given class:

$$e_{i,c} = (n_{i,c}+1)^{\gamma}\mathbf{1}[n_{i,c} > 0], \tag{2}$$

where $\gamma = 0.55$ is the default evidence exponent. This weight gives larger class-specific contribution to clients with more supervised samples of class c, while excluding clients that never observe this class. The normalized reliability is

$$\alpha_{i,c} = \frac{e_{i,c}}{\sum_{j=1}^{K} e_{j,c}}. \tag{3}$$

The global diagnosis prototype is obtained by class-wise reliability-weighted aggregation:

$$p_c = \sum_{i=1}^{K} \alpha_{i,c}\mu_{i,c}. \tag{4}$$

This step aggregates evidence only from clients that have actually observed class c, preventing absent-class classifiers from diluting valid diagnostic signals.

(3) Prototype-induced Diagnostic Head. Finally, LAMP-Merge maps global diagnosis prototypes into a cosine prototype classifier:

$$w_c = s\frac{p_c}{\|p_c\|_2}, \tag{5}$$

where $s = 18.75$ is the default head scale. The resulting head is not an average of local classifier parameters; it is induced by class evidence in the shared reference space, thereby restoring multiclass diagnostic directions over the global label space.

Long-tail Prevalence Calibration

To address **long-tailed medical prevalence cannot be ignored**, LPC adds a bounded class prior after DPR to calibrate the clinical prevalence reflected by client statistics.

(1) Client-reported Prevalence Prior. First, each client uploads class-prevalence counts $m_{i,c}$. The server estimates the global class prior

$$\pi_c = \frac{\sum_{i=1}^{K} m_{i,c}}{\sum_{k=1}^{C}\sum_{i=1}^{K} m_{i,k}}. \tag{6}$$

This prior is computed only from class-level counts, not from raw images, per-sample features, or per-sample predictions.

(2) Long-tail Activation Criterion. Second, to avoid unnecessary calibration in non-long-tailed cases, the server measures dominant-class imbalance by

$$r = C \max_c \pi_c. \tag{7}$$

The calibration module is activated only when $r > \tau$, where $\tau = 2.5$ by default.

(3) Centered Log-prior Calibration. Finally, when the module is activated, LAMP-Merge constructs a centered log-prior bias

$$b_c = \lambda\left(\log \pi_c - \frac{1}{C}\sum_{k=1}^{C}\log \pi_k\right), \tag{8}$$

with default maximum calibration strength $\lambda = 4.25$. The centering term prevents a uniform shift of all class logits, so the bias only changes the relative class prior. This module calibrates prevalence without replacing the diagnostic directions reconstructed by the first module.

Optimization Objective and Algorithm

The final scoring function of LAMP-Merge integrates the diagnostic prototype reconstruction term and the long-tail prevalence calibration term:

$$\ell_c(x) = w_c^{\top}\phi_0(T(x)) + \mathbf{1}[r>\tau]\, b_c. \tag{9}$$

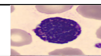
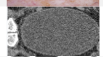
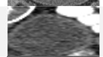
Experiments

Experimental Settings

(1) Evaluation Datasets: We evaluate the merged models on five representative MedMNIST v2 benchmarks(Yang, Shi, and Ni 2023): Blood, Derma, Organ-C, Organ-S, and Ultrasound. They cover blood-cell microscopy, dermoscopy, abdominal CT, and ultrasound image classification, and together span different class counts and prevalence profiles.

(2) Baselines: We compare LAMP-Merge with twelve established merging baselines spanning parameter averaging, task-vector pruning, conflict resolution, statistic-aware correction, interpolation, subspace merging, Fourier fusion, and robust merging (Yadav et al. 2023; Yu et al. 2024; Jin et al. 2025; Wu et al. 2025; Davari and Belilovsky 2024; Jang, Yun, and Han 2025; Ilharco et al. 2023; Marczak et al. 2025; Zheng and Wang 2025; Zeng et al. 2025).

Table 1: Dataset statistics and representative samples for the five medical image benchmarks. Counts are taken from the exact NPZ files used in our experiments.

Dataset	Classes	Train	Val	Test	Total	Sample
Blood	8	11,959	1,712	3,421	17,092	
Derma	7	7,007	1,003	2,005	10,015	
Organ-C	11	12,975	2,392	8,216	23,583	
Organ-S	11	13,932	2,452	8,827	25,211	
Ultrasound	8	3,869	549	1,113	5,531	

(3) Merging Settings: We evaluate four vision-model families: ResNet, ConvNeXt, ViT-Tiny, and Swin-Tiny(He et al. 2016; Liu et al. 2022; Dosovitskiy et al. 2021; Liu et al. 2021). For each dataset-model pair, we construct non-IID client partitions with $K \in \{3, 5, 7\}$ clients and Dirichlet skew parameter $\beta \in \{0, 0.01, 0.1\}$, using seed 42(Hsu, Qi, and Brown 2019). A client-average result fixes the dataset, model, and K, and then averages over the three β values. And the experiments follow a single-shot asynchronous model-merging protocol.

Comparison With the State of the Arts

Tables 2, 3, 4, and 5 report the full client-average results for all four backbones. Across all evaluated settings, LAMP-Merge wins or ties the strongest non-LAMP baseline in **76 cells (95.0%)**.

Ablation Study

To assess DPR and LPC, we conduct module ablations under the main experimental configuration. Weight averaging, weight averaging + DPR, and full LAMP-Merge obtain mean ACC of 22.04%, 58.91%, and 62.21%, respectively; Fig. 4 further reports the results that use TIES/DARE as baselines.

1) Effectiveness of DPR To identify the necessary components of DPR, we retain LPC and probe two design axes. The first axis alters the *prototype semantics*: classifier-head aggregation (CHA) swaps each class prototype for the local classifier-head direction, testing whether independently trained heads already share a class-aligned representation; global-feature mean (GFM) reuses one class-agnostic feature mean for every class, testing whether generic client appearance alone can drive merging; support-only synthetic head (SSH) keeps the cosine-head interface but fills it with a fixed random unit vector, testing whether the head form or support counts suffice without diagnostic content; and shuffled-label prototype (SLP) permutes prototypes across labels, testing whether class evidence must be assigned to the correct output class. The second axis alters the *client-support weighting*: uniform client weight (UCW) credits every observing client equally regardless of sample count; binary support only (BSO) collapses both support and prevalence to presence indicators, removing all frequency information; and global client-size weight (GCSW) weights by a client's total size rather than its class-specific support. These probe how reliably each observed class should be credited.

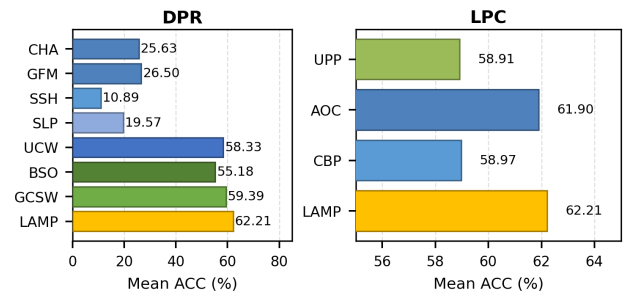


Figure 3: Internal ablations of DPR and LPC. The plotted abbreviations are defined in the ablation text above.

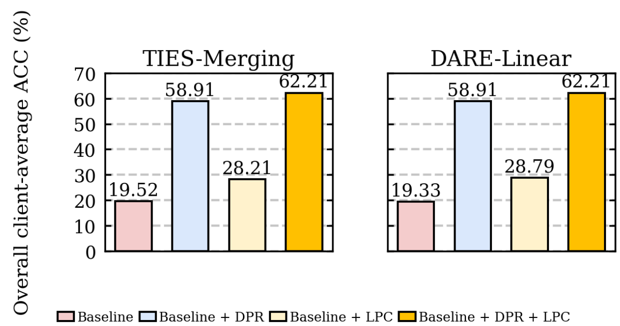


Figure 4: Strict 2 × 2 DPR/LPC ablations for TIES-Merging and DARE-Linear. **All experimental datasets are included.**

Table 6 and Fig. 3 show that every replacement lowers ACC. This confirms that DPR needs both reliable class prototypes and class-specific support weighting.

2) Effectiveness of LPC To evaluate LPC, we retain DPR and compare its empirical prior against three controls across the full experimental setting. Uniform prevalence prior (UPP) replaces the measured class distribution with a flat prior, testing whether calibration should ignore observed prevalence altogether. Always-on calibration (AOC) removes the long-tail activation gate so the prior is applied to every configuration, testing whether the gate is a useful safeguard against calibrating near-balanced cases. Client-balanced prior (CBP) estimates the prior by averaging each client's normalized prevalence with equal weight, testing whether equal client influence can replace a sample-count-weighted prevalence estimate.

Table 7 and Fig. 3 show that LAMP-Merge is best and every control reduces ACC. This demonstrates that calibrated empirical prevalence and client-size-aware aggregation complement DPR.

Table 2: Client-average test accuracy / Macro-F1 (%) on Swin-Tiny. Each entry is reported as ACC / F1, where F1 denotes Macro-F1 and each value averages the three Dirichlet settings for a fixed client number; Avg averages $K = 3, 5, 7$ within the corresponding dataset.

Method	Blood				Derma				Organ-C				Organ-S				Ultrasound			
	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg
Weight Avg.[ICML 2022]	15.65 / 3.37	11.86 / 2.72	15.24 / 3.44	14.25 / 3.18	45.69 / 7.94	48.30 / 8.59	66.88 / 11.45	53.62 / 9.33	10.68 / 3.45	12.94 / 2.45	7.87 / 1.53	10.50 / 2.48	6.82 / 1.45	6.04 / 1.51	13.81 / 2.31	8.89 / 1.76	15.27 / 3.42	12.94 / 2.89	14.65 / 3.23	14.29 / 3.18
TIES[NeurIPS 2023]	9.72 / 2.20	11.90 / 2.61	13.84 / 3.00	11.82 / 2.60	8.41 / 2.19	7.80 / 2.01	66.88 / 11.45	27.70 / 5.22	11.61 / 3.92	7.02 / 1.40	9.34 / 1.92	9.32 / 2.41	7.34 / 2.39	6.06 / 1.10	11.08 / 2.28	8.16 / 1.92	16.95 / 5.90	13.03 / 2.98	12.61 / 3.51	14.20 / 4.13
DARE-Linear[ICML 2024]	15.65 / 3.37	11.61 / 2.94	18.25 / 5.30	15.17 / 3.87	45.69 / 7.94	27.71 / 5.24	46.30 / 8.10	39.90 / 7.09	11.55 / 4.05	12.72 / 1.99	7.76 / 1.39	10.68 / 2.48	6.40 / 1.29	7.20 / 1.92	13.81 / 2.79	9.14 / 2.00	17.43 / 3.83	15.36 / 5.44	14.97 / 3.39	15.92 / 4.22
DARE-TIES[ICML 2024]	13.84 / 3.00	11.62 / 2.55	13.84 / 3.00	13.10 / 2.85	29.61 / 5.70	26.38 / 4.88	66.88 / 11.45	40.96 / 7.34	12.76 / 4.00	7.74 / 1.30	11.16 / 2.82	10.55 / 2.71	11.94 / 5.42	4.34 / 0.79	12.30 / 2.82	9.53 / 3.01	17.91 / 4.84	13.57 / 3.14	13.03 / 3.02	14.83 / 3.66
RegMean[ICLR 2023]	13.74 / 3.02	17.15 / 3.65	15.09 / 3.43	15.33 / 3.37	27.05 / 5.06	46.30 / 8.10	45.69 / 7.94	39.68 / 7.03	14.52 / 6.71	7.73 / 1.30	8.11 / 1.36	10.12 / 3.12	10.45 / 3.40	6.74 / 1.58	8.50 / 1.90	8.57 / 2.29	16.95 / 6.19	10.90 / 4.14	18.60 / 8.55	15.48 / 6.29
Fisher[NeurIPS 2022]	15.65 / 3.37	17.56 / 3.72	15.65 / 3.37	16.29 / 3.49	65.70 / 11.46	6.52 / 1.72	42.71 / 8.23	38.31 / 7.14	8.31 / 3.23	9.01 / 1.49	10.14 / 3.19	9.15 / 2.64	6.95 / 1.82	8.29 / 2.40	12.56 / 4.10	9.27 / 2.77	17.22 / 6.16	13.51 / 3.60	16.62 / 4.13	15.78 / 4.63
Breadcrumbs[ECCV 2024]	15.00 / 3.57	13.65 / 2.97	15.37 / 3.92	14.67 / 3.48	6.47 / 1.71	27.71 / 5.24	7.08 / 3.83	13.75 / 3.59	14.44 / 5.14	17.83 / 2.81	16.60 / 2.51	16.29 / 3.49	23.54 / 3.46	20.77 / 3.11	21.11 / 3.57	21.81 / 3.38	14.47 / 5.23	13.12 / 2.89	16.53 / 4.71	14.71 / 4.28
Model Stock[ECCV 2024]	13.74 / 3.02	14.40 / 3.55	17.15 / 3.65	15.10 / 3.41	45.69 / 7.94	27.71 / 5.24	48.25 / 8.79	40.55 / 7.32	11.10 / 3.38	7.71 / 1.49	14.04 / 2.91	10.95 / 2.59	6.71 / 1.43	11.06 / 3.00	17.13 / 4.69	11.63 / 3.04	15.39 / 3.60	12.73 / 3.24	18.72 / 7.22	15.61 / 4.69
Iso-C[ICML 2025]	14.30 / 6.44	17.22 / 7.05	17.00 / 9.99	16.17 / 7.83	31.01 / 6.61	28.38 / 8.12	51.74 / 11.15	37.04 / 8.63	13.02 / 7.20	13.27 / 5.57	14.70 / 7.16	13.67 / 6.64	15.45 / 7.27	17.26 / 7.50	18.80 / 9.90	17.17 / 8.22	17.55 / 7.09	17.34 / 8.58	17.91 / 7.02	17.60 / 7.56
Free-Merging[ICCV 2025]	11.54 / 2.57	8.61 / 2.54	18.40 / 6.97	12.85 / 4.03	45.69 / 7.94	48.30 / 8.59	66.88 / 11.45	53.62 / 9.33	9.59 / 2.25	13.77 / 3.37	8.90 / 2.00	10.75 / 2.54	6.82 / 1.42	4.97 / 0.86	10.62 / 1.73	7.47 / 1.34	15.18 / 3.28	15.75 / 5.69	15.99 / 5.13	15.64 / 4.70
RobustMerge[NeurIPS 2025]	13.74 / 3.02	14.09 / 3.14	16.70 / 3.60	14.84 / 3.26	23.77 / 4.23	27.71 / 5.24	5.04 / 1.46	18.84 / 3.64	16.01 / 4.91	14.42 / 2.35	19.60 / 3.97	16.68 / 3.74	16.73 / 4.92	11.51 / 3.64	20.77 / 3.11	16.33 / 3.89	17.61 / 6.32	13.15 / 3.03	16.29 / 5.38	15.68 / 4.91
FROM[AAAI 2026]	15.65 / 3.37	14.95 / 3.20	16.71 / 3.57	15.77 / 3.38	45.69 / 7.94	29.71 / 5.72	66.88 / 11.45	47.43 / 8.37	11.74 / 3.84	12.20 / 1.91	10.51 / 2.83	11.48 / 2.86	11.32 / 4.25	10.16 / 3.16	13.80 / 2.20	11.76 / 3.20	15.39 / 3.79	12.52 / 2.77	14.76 / 5.38	14.23 / 3.98
LAMP-Merge	**77.51 / 75.93**	**77.43 / 75.77**	**77.25 / 75.61**	**77.40 / 75.77**	**63.26 / 28.08**	**63.13 / 28.23**	**63.13 / 27.54**	**63.17 / 27.95**	**68.47 / 65.83**	**68.17 / 65.38**	**68.16 / 65.32**	**68.27 / 65.51**	**62.60 / 58.23**	**62.79 / 58.45**	**62.83 / 58.37**	**62.74 / 58.35**	**47.68 / 45.72**	**48.61 / 46.76**	**48.28 / 46.46**	**48.19 / 46.31**
	+61.86 / +69.49	**+59.87 / +68.72**	**+58.85 / +65.62**	**+61.11 / +67.94**	**-2.44 / +16.62**	**+14.83 / +19.64**	**-3.75 / +16.09**	**+9.55 / +18.62**	**+52.46 / +58.63**	**+50.34 / +59.81**	**+48.56 / +58.16**	**+51.59 / +58.87**	**+39.06 / +50.96**	**+42.02 / +50.95**	**+41.72 / +48.47**	**+40.93 / +50.13**	**+29.77 / +38.63**	**+31.27 / +38.18**	**+29.56 / +37.91**	**+30.59 / +38.75**

Table 3: Client-average test accuracy / Macro-F1 (%) on ConvNeXt. Each entry is reported as ACC / F1, where F1 denotes Macro-F1 and each value averages the three Dirichlet settings for a fixed client number; Avg averages $K = 3, 5, 7$ within the corresponding dataset.

Method	Blood				Derma				Organ-C				Organ-S				Ultrasound			
	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg
Weight Avg.[ICML 2022]	15.65 / 3.37	11.22 / 2.48	13.84 / 3.00	13.57 / 2.95	29.66 / 5.71	48.30 / 8.59	66.88 / 11.45	48.28 / 8.58	10.37 / 2.79	12.18 / 1.91	9.44 / 1.57	10.66 / 2.09	13.08 / 2.04	10.35 / 1.69	14.16 / 2.20	12.53 / 1.98	15.06 / 3.26	15.06 / 3.26	12.52 / 2.77	14.22 / 3.09
TIES[NeurIPS 2023]	17.15 / 3.65	12.14 / 2.67	15.24 / 3.30	14.84 / 3.21	29.66 / 5.71	18.00 / 5.62	27.12 / 5.12	24.93 / 5.48	12.17 / 4.98	7.86 / 1.43	7.16 / 1.61	9.06 / 2.67	11.23 / 3.07	7.61 / 2.02	7.96 / 1.40	8.93 / 2.16	15.18 / 3.28	14.56 / 3.15	12.91 / 2.84	14.22 / 3.09
DARE-Linear[ICML 2024]	17.56 / 3.76	11.22 / 2.48	13.45 / 2.92	14.07 / 3.05	27.05 / 5.06	48.30 / 8.59	48.30 / 8.59	41.21 / 7.41	10.35 / 2.40	6.99 / 1.18	10.85 / 1.78	9.39 / 1.79	12.82 / 2.16	10.35 / 1.69	7.97 / 1.33	10.38 / 1.73	15.75 / 3.39	15.75 / 3.39	12.64 / 2.79	14.71 / 3.19
DARE-TIES[ICML 2024]	10.96 / 3.30	15.19 / 3.27	14.97 / 3.66	13.71 / 3.41	9.08 / 2.36	27.98 / 7.91	24.51 / 4.47	20.52 / 4.91	13.04 / 4.57	13.46 / 3.30	7.70 / 1.47	11.40 / 3.11	10.97 / 3.43	9.46 / 2.64	7.09 / 1.58	9.17 / 2.55	14.56 / 3.15	12.97 / 2.85	15.00 / 3.26	14.18 / 3.09
RegMean[ICLR 2023]	15.65 / 3.37	8.17 / 1.89	18.81 / 5.87	14.21 / 3.71	26.33 / 4.87	7.78 / 2.03	27.10 / 5.07	20.40 / 3.99	11.56 / 3.99	10.46 / 2.43	7.18 / 1.64	9.73 / 2.68	10.09 / 4.29	12.34 / 4.42	10.49 / 3.14	10.97 / 3.95	15.75 / 3.39	15.39 / 3.33	13.24 / 2.91	14.79 / 3.21
Fisher[NeurIPS 2022]	15.65 / 3.37	9.99 / 2.26	11.93 / 2.71	12.52 / 2.78	48.25 / 8.58	26.38 / 4.88	29.71 / 5.72	34.78 / 6.39	10.50 / 3.63	8.69 / 2.45	8.60 / 1.44	9.26 / 2.51	6.68 / 2.23	4.67 / 0.99	12.27 / 3.15	7.87 / 2.13	16.95 / 3.62	15.75 / 3.39	14.56 / 3.15	15.75 / 3.39
Breadcrumbs[ECCV 2024]	15.16 / 3.26	13.66 / 2.97	11.63 / 2.55	13.49 / 2.93	5.24 / 1.38	27.66 / 5.22	46.30 / 8.10	26.40 / 4.90	10.31 / 1.70	14.44 / 2.35	15.75 / 3.09	13.50 / 2.38	15.16 / 2.35	15.54 / 2.40	20.77 / 3.11	17.15 / 2.62	13.84 / 3.03	13.63 / 2.99	13.63 / 2.99	13.70 / 3.00
Model Stock[ECCV 2024]	17.56 / 3.72	11.44 / 2.54	17.78 / 3.80	15.59 / 3.35	29.66 / 5.71	48.30 / 8.59	66.88 / 11.45	48.28 / 8.58	12.76 / 3.72	12.18 / 1.91	13.85 / 2.17	12.93 / 2.60	13.08 / 2.04	10.35 / 1.69	19.35 / 2.91	14.26 / 2.21	15.75 / 3.39	12.91 / 2.84	14.79 / 3.21	14.49 / 3.15
Iso-C[ICML 2025]	19.47 / 4.07	18.65 / 3.93	14.80 / 3.22	17.64 / 3.74	29.71 / 5.72	48.30 / 8.59	66.88 / 11.45	48.30 / 8.59	9.91 / 2.18	6.99 / 1.18	9.45 / 1.65	8.78 / 1.67	7.89 / 1.33	14.77 / 2.28	8.66 / 3.00	10.44 / 2.20	15.18 / 3.28	12.91 / 2.84	16.05 / 3.50	14.71 / 3.21
Free-Merging[ICCV 2025]	15.75 / 3.35	11.90 / 2.61	13.84 / 3.00	13.83 / 2.99	29.71 / 5.72	48.30 / 8.59	66.88 / 11.45	48.30 / 8.59	9.06 / 1.55	12.18 / 1.91	13.85 / 2.17	11.69 / 1.87	13.08 / 2.04	10.35 / 1.69	14.16 / 2.20	12.53 / 1.98	15.06 / 3.26	12.91 / 2.84	16.95 / 3.92	14.97 / 3.34
RobustMerge[NeurIPS 2025]	14.10 / 3.06	14.05 / 3.05	7.52 / 1.75	11.89 / 2.62	9.08 / 2.36	45.07 / 7.77	48.30 / 8.59	34.15 / 6.24	13.94 / 2.34	12.18 / 1.91	17.93 / 2.72	14.68 / 2.32	23.54 / 3.47	9.61 / 1.57	20.77 / 3.11	17.97 / 2.72	13.00 / 2.86	11.35 / 2.55	13.63 / 2.99	12.66 / 2.80
FROM[AAAI 2026]	15.65 / 3.37	10.83 / 2.39	16.71 / 3.57	14.39 / 3.11	29.66 / 5.71	34.08 / 8.52	45.69 / 7.94	36.48 / 7.39	12.32 / 3.83	12.24 / 1.92	6.50 / 1.10	10.35 / 2.28	13.24 / 3.47	11.05 / 2.64	8.97 / 1.49	11.09 / 2.54	15.75 / 3.39	15.78 / 3.40	10.87 / 2.45	14.14 / 3.08
LAMP-Merge	**85.00 / 83.43**	**85.04 / 83.47**	**85.11 / 83.54**	**85.05 / 83.48**	**59.67 / 33.64**	**58.79 / 32.92**	**59.50 / 33.42**	**59.32 / 33.33**	**65.40 / 63.17**	**65.31 / 63.11**	**65.49 / 63.23**	**65.40 / 63.17**	**60.53 / 56.03**	**60.48 / 56.04**	**60.62 / 56.04**	**60.55 / 56.04**	**43.01 / 41.00**	**42.83 / 40.79**	**42.35 / 40.36**	**42.73 / 40.72**
	+65.53 / +79.36	**+66.39 / +79.54**	**+66.30 / +77.67**	**+67.41 / +79.74**	**+11.42 / +25.06**	**+10.49 / +24.33**	**-7.38 / +21.97**	**+11.02 / +24.74**	**+51.46 / +58.19**	**+50.87 / +59.81**	**+47.56 / +60.14**	**+50.72 / +60.06**	**+36.99 / +51.74**	**+44.94 / +51.62**	**+39.85 / +52.89**	**+42.58 / +52.09**	**+26.06 / +37.38**	**+27.05 / +37.39**	**+25.40 / +36.44**	**+26.98 / +37.33**

Table 4: Client-average test accuracy / Macro-F1 (%) on ResNet. Each entry is reported as ACC / F1, where F1 denotes Macro-F1 and each value averages the three Dirichlet settings for a fixed client number; Avg averages $K = 3, 5, 7$ within the corresponding dataset.

Method	Blood				Derma				Organ-C				Organ-S				Ultrasound			
	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg
Weight Avg.[ICML 2022]	33.45 / 23.94	24.66 / 15.61	28.21 / 14.88	28.77 / 18.14	62.91 / 13.63	47.00 / 11.83	62.44 / 13.44	57.45 / 12.97	22.61 / 12.19	17.71 / 9.10	13.96 / 6.49	18.10 / 9.26	28.64 / 16.12	17.94 / 9.68	24.75 / 9.13	23.77 / 11.64	24.35 / 12.05	15.81 / 6.89	15.42 / 4.50	18.53 / 7.81
TIES[NeurIPS 2023]	26.01 / 15.52	18.72 / 6.45	17.87 / 6.48	20.86 / 9.48	59.95 / 17.19	48.41 / 9.06	64.84 / 12.59	57.73 / 12.95	26.50 / 14.04	16.84 / 10.62	21.18 / 9.84	21.51 / 11.50	19.81 / 10.76	19.15 / 10.56	18.73 / 6.21	19.23 / 9.18	34.77 / 25.99	19.89 / 11.44	18.54 / 7.58	24.40 / 15.00
DARE-Linear[ICML 2024]	28.31 / 18.65	28.56 / 15.87	23.92 / 10.84	26.93 / 15.12	63.57 / 14.49	48.43 / 11.06	62.88 / 13.03	58.29 / 12.86	20.18 / 8.37	25.88 / 10.20	16.50 / 6.84	20.85 / 8.47	25.45 / 11.52	22.45 / 9.05	25.37 / 8.26	24.42 / 9.61	21.62 / 9.40	17.28 / 8.28	16.08 / 4.93	18.33 / 7.53
DARE-TIES[ICML 2024]	31.95 / 19.01	19.36 / 6.63	17.80 / 7.03	23.04 / 10.89	53.75 / 11.16	48.25 / 8.70	55.58 / 10.94	52.52 / 10.27	18.51 / 7.52	15.15 / 7.22	18.59 / 7.54	17.42 / 7.43	17.90 / 10.39	13.52 / 6.84	17.30 / 7.61	16.24 / 8.28	29.32 / 20.58	16.32 / 8.63	18.90 / 7.67	21.51 / 12.29
RegMean[ICLR 2023]	33.09 / 25.73	27.17 / 17.43	30.97 / 19.68	30.41 / 20.95	60.17 / 16.28	28.45 / 14.49	45.99 / 15.32	44.87 / 15.36	24.39 / 15.91	13.54 / 11.81	12.68 / 7.91	16.87 / 11.88	26.04 / 16.75	16.39 / 12.52	11.98 / 7.07	18.14 / 12.11	26.21 / 17.86	20.93 / 15.76	20.10 / 15.47	22.41 / 16.36
Fisher[NeurIPS 2022]	37.11 / 26.20	24.77 / 13.84	17.69 / 6.73	26.53 / 15.59	67.02 / 16.51	67.17 / 12.67	66.90 / 11.54	67.03 / 13.58	32.38 / 22.82	33.29 / 23.98	25.31 / 11.32	30.33 / 19.37	26.55 / 16.73	30.44 / 17.28	18.97 / 9.70	25.32 / 14.57	27.16 / 18.98	27.91 / 18.22	20.69 / 11.18	25.26 / 16.13
Breadcrumbs[ECCV 2024]	8.53 / 3.18	14.17 / 5.35	14.60 / 6.13	12.43 / 4.89	4.95 / 1.94	29.13 / 7.30	17.21 / 5.39	17.10 / 4.88	7.66 / 1.79	6.76 / 2.68	6.85 / 2.60	7.09 / 2.36	8.02 / 2.73	9.48 / 3.68	6.03 / 1.91	7.85 / 2.77	12.25 / 3.32	12.73 / 5.09	10.21 / 5.50	11.73 / 4.64
Model Stock[ECCV 2024]	24.20 / 16.77	18.23 / 8.12	19.75 / 6.48	20.73 / 10.45	62.18 / 14.60	43.11 / 10.80	57.22 / 13.08	54.17 / 12.83	12.25 / 3.93	12.74 / 6.03	11.85 / 3.01	12.28 / 4.32	27.43 / 14.71	15.79 / 7.86	20.12 / 5.99	21.11 / 9.52	17.91 / 8.10	13.21 / 5.94	15.78 / 4.44	15.63 / 6.16
Iso-C[ICML 2025]	32.78 / 23.32	27.48 / 17.39	30.23 / 15.01	30.16 / 18.57	50.49 / 15.02	44.22 / 16.17	64.16 / 14.32	52.96 / 15.17	23.31 / 12.98	16.29 / 7.96	13.83 / 6.14	17.81 / 9.03	27.36 / 14.97	21.18 / 11.80	24.53 / 8.07	24.36 / 11.61	26.15 / 15.77	16.98 / 8.93	15.54 / 5.75	19.56 / 10.15
Free-Merging[ICCV 2025]	33.45 / 23.95	24.68 / 15.63	28.21 / 14.89	28.78 / 18.16	62.89 / 13.62	47.00 / 11.83	62.48 / 13.45	57.46 / 12.97	22.61 / 12.19	17.71 / 9.10	13.96 / 6.49	18.10 / 9.26	28.63 / 16.12	17.94 / 9.68	24.75 / 9.12	23.77 / 11.64	24.35 / 12.05	15.81 / 6.88	15.45 / 4.51	18.54 / 7.81
RobustMerge[NeurIPS 2025]	34.01 / 20.06	24.30 / 13.12	22.56 / 9.35	26.95 / 14.18	29.56 / 11.45	20.85 / 9.33	39.42 / 14.11	29.94 / 11.63	24.58 / 13.59	17.64 / 9.42	16.54 / 6.77	19.59 / 9.93	26.80 / 12.52	22.28 / 13.00	23.01 / 9.31	24.03 / 11.61	21.89 / 11.26	22.31 / 12.73	17.07 / 7.26	20.43 / 10.42
FROM[AAAI 2026]	29.89 / 18.76	23.06 / 11.03	20.52 / 8.00	24.49 / 12.60	67.13 / 13.11	43.69 / 10.22	65.80 / 12.95	58.88 / 12.09	23.50 / 13.76	15.60 / 9.50	15.20 / 5.57	18.10 / 9.61	22.07 / 12.49	20.87 / 11.30	24.76 / 8.15	22.57 / 10.64	23.09 / 11.06	15.06 / 7.44	20.63 / 8.82	19.60 / 9.11
LAMP-Merge	**80.06 / 78.72**	**79.99 / 78.67**	**79.99 / 78.65**	**80.01 / 78.68**	**67.55 / 15.03**	**67.63 / 15.16**	**67.63 / 14.94**	**67.60 / 15.04**	**60.08 / 53.15**	**59.98 / 52.98**	**60.07 / 53.28**	**60.04 / 53.14**	**52.98 / 42.85**	**52.80 / 42.54**	**52.82 / 42.31**	**52.87 / 42.57**	**48.07 / 45.96**	**46.72 / 44.46**	**47.38 / 45.08**	**47.39 / 45.17**
	+42.95 / +52.52	**+51.43 / +61.24**	**+49.02 / +58.97**	**+49.60 / +57.73**	**+0.42 / -2.16**	**+0.46 / -1.01**	**+0.73 / -0.38**	**+0.57 / -0.32**	**+27.70 / +30.33**	**+26.69 / +29.00**	**+34.76 / +41.96**	**+29.71 / +33.77**	**+24.34 / +26.10**	**+22.36 / +25.26**	**+27.45 / +32.61**	**+27.55 / +28.00**	**+13.30 / +19.97**	**+18.81 / +26.24**	**+26.69 / +29.61**	**+22.13 / +28.81**

Table 5: Client-average test accuracy / Macro-F1 (%) on ViT-Tiny. Each entry is reported as ACC / F1, where F1 denotes Macro-F1 and each value averages the three Dirichlet settings for a fixed client number; Avg averages $K = 3, 5, 7$ within the corresponding dataset.

Method	Blood				Derma				Organ-C				Organ-S				Ultrasound			
	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg
Weight Avg.[ICML 2022]	13.45 / 4.05	10.82 / 2.39	11.60 / 3.15	11.96 / 3.19	46.38 / 8.14	43.69 / 11.46	36.96 / 8.44	42.34 / 9.35	16.57 / 7.59	10.43 / 3.88	9.52 / 4.01	12.18 / 5.16	8.44 / 3.70	8.52 / 2.91	17.10 / 5.24	11.35 / 3.95	14.50 / 3.58	13.24 / 3.56	18.84 / 7.56	15.52 / 4.90
TIES[NeurIPS 2023]	7.96 / 3.42	14.21 / 5.13	12.85 / 3.43	11.67 / 3.99	47.80 / 9.19	34.98 / 8.17	46.15 / 8.46	42.98 / 8.61	13.40 / 6.55	12.24 / 4.69	9.53 / 3.40	11.72 / 4.88	11.92 / 3.97	10.02 / 2.66	15.97 / 5.94	12.64 / 4.19	20.01 / 9.29	13.06 / 4.93	10.18 / 2.58	14.42 / 5.60
DARE-Linear[ICML 2024]	23.96 / 12.03	11.31 / 3.46	13.87 / 4.27	16.38 / 6.59	28.51 / 5.74	10.02 / 3.26	12.04 / 5.72	16.86 / 4.91	15.33 / 8.12	9.46 / 2.88	7.04 / 2.83	10.61 / 4.61	7.48 / 3.58	5.49 / 1.28	16.10 / 5.28	9.69 / 3.38	13.66 / 3.89	10.51 / 2.38	14.35 / 3.21	12.84 / 3.16
DARE-TIES[ICML 2024]	7.35 / 2.90	11.89 / 2.61	11.46 / 2.81	10.23 / 2.77	27.35 / 6.66	45.02 / 9.40	35.26 / 8.36	35.88 / 8.14	12.52 / 5.80	9.03 / 4.08	10.33 / 3.58	10.63 / 4.49	5.41 / 1.64	8.74 / 2.73	5.91 / 2.50	6.69 / 2.29	14.47 / 5.16	11.17 / 3.72	10.42 / 3.00	12.02 / 3.96
RegMean[ICLR 2023]	17.92 / 8.00	16.26 / 6.17	13.17 / 2.95	15.78 / 5.71	51.17 / 10.12	48.18 / 8.86	66.75 / 11.44	55.37 / 10.14	12.60 / 4.50	9.61 / 3.44	9.25 / 4.00	10.49 / 3.98	13.23 / 5.96	7.36 / 2.21	10.45 / 2.68	10.35 / 3.62	15.09 / 5.37	16.35 / 6.75	15.57 / 6.60	15.67 / 6.24
Fisher[NeurIPS 2022]	9.43 / 2.74	14.12 / 3.09	19.65 / 7.18	14.40 / 4.34	48.53 / 8.99	24.67 / 7.27	48.16 / 9.57	40.45 / 8.61	11.95 / 5.93	9.21 / 3.80	9.17 / 3.58	10.11 / 4.44	9.20 / 4.06	13.89 / 4.07	13.07 / 5.92	12.05 / 4.68	20.84 / 9.45	12.37 / 5.22	14.17 / 3.20	15.79 / 5.96
Breadcrumbs[ECCV 2024]	13.64 / 3.67	10.39 / 2.34	13.00 / 3.72	12.34 / 3.24	4.89 / 1.99	25.72 / 4.75	6.07 / 3.14	12.22 / 3.29	7.73 / 2.44	10.01 / 5.62	16.44 / 4.60	11.39 / 4.22	12.37 / 4.42	4.83 / 1.76	12.19 / 3.51	9.80 / 3.23	13.60 / 4.96	11.41 / 2.71	15.24 / 3.87	13.42 / 3.85
Model Stock[ECCV 2024]	19.41 / 6.62	11.21 / 3.74	11.33 / 3.14	13.98 / 4.50	38.05 / 7.67	30.81 / 7.56	27.35 / 7.04	32.07 / 7.42	14.88 / 7.82	12.27 / 6.18	12.16 / 4.29	13.10 / 6.10	8.93 / 3.35	8.59 / 3.27	18.65 / 6.91	12.06 / 4.51	15.99 / 5.30	18.09 / 6.89	16.41 / 4.46	16.83 / 5.55
Iso-C[ICML 2025]	12.36 / 3.63	14.83 / 5.69	10.19 / 3.41	12.46 / 4.24	57.11 / 11.23	45.30 / 11.31	23.04 / 7.63	41.82 / 10.06	10.84 / 4.97	5.67 / 1.78	8.97 / 5.53	8.49 / 4.09	6.12 / 2.27	5.32 / 2.29	10.81 / 5.45	7.42 / 3.34	14.17 / 5.75	15.21 / 6.49	18.72 / 8.56	16.03 / 6.93
Free-Merging[ICCV 2025]	19.89 / 7.03	14.57 / 6.11	13.92 / 5.51	16.13 / 6.22	46.30 / 8.10	47.70 / 9.56	38.59 / 9.31	44.20 / 8.99	14.25 / 5.22	9.80 / 4.05	7.58 / 3.26	10.54 / 4.18	7.27 / 2.91	9.13 / 3.42	16.43 / 4.71	10.94 / 3.68	14.05 / 3.22	15.93 / 4.71	16.47 / 6.19	15.48 / 4.71
RobustMerge[NeurIPS 2025]	14.82 / 3.29	10.70 / 3.15	14.25 / 3.47	13.25 / 3.30	3.46 / 1.10	27.68 / 5.27	16.77 / 8.45	15.97 / 4.94	9.54 / 3.90	6.17 / 2.08	14.37 / 5.33	10.03 / 3.77	12.60 / 3.27	7.08 / 4.07	12.17 / 2.61	10.62 / 3.32	17.01 / 6.77	18.54 / 7.33	15.63 / 4.37	17.06 / 6.16
FROM[AAAI 2026]	18.01 / 8.54	7.81 / 2.36	14.05 / 3.05	13.29 / 4.65	46.95 / 9.16	51.21 / 10.76	66.88 / 11.45	55.01 / 10.46	21.48 / 10.56	13.00 / 5.13	9.72 / 3.53	14.73 / 6.41	7.72 / 3.10	7.97 / 2.50	14.86 / 6.23	10.18 / 3.94	18.36 / 7.49	17.43 / 7.51	17.79 / 8.40	17.86 / 7.80
LAMP-Merge	**85.30 / 83.06**	**85.19 / 82.96**	**85.15 / 82.85**	**85.21 / 82.95**	**59.47 / 36.65**	**59.38 / 36.61**	**59.80 / 36.79**	**59.55 / 36.68**	**58.73 / 57.39**	**58.39 / 56.93**	**58.28 / 56.80**	**58.47 / 57.04**	**54.36 / 52.06**	**54.47 / 52.22**	**54.60 / 52.28**	**54.48 / 52.18**	**46.09 / 42.27**	**46.03 / 42.23**	**45.88 / 42.15**	**46.00 / 42.22**
	+61.34 / +71.03	**+68.93 / +76.79**	**+65.50 / +75.67**	**+68.83 / +76.36**	**+2.36 / +25.42**	**+8.17 / +25.15**	**-7.08 / +25.34**	**+4.18 / +26.22**	**+37.25 / +46.83**	**+45.39 / +50.75**	**+41.84 / +51.27**	**+43.74 / +50.63**	**+41.13 / +46.10**	**+40.58 / +48.15**	**+35.95 / +45.37**	**+41.84 / +47.50**	**+25.25 / +32.82**	**+27.49 / +34.72**	**+27.04 / +33.59**	**+28.14 / +34.42**

Innovative prediction-collapse Diagnostics

Because raw ACC can hide single-class predictions on long-tailed data, we additionally report collapse ratio, effective predicted classes, and predicted–true total variation(Buda, Maki, and Mazurowski 2018; Brodersen et al. 2010). With q_c and $\bar{q}_c$ the predicted and true class distributions over the test set, the three diagnostics are

$$\rho = \max_c q_c, \qquad C_{\text{eff}} = \exp\left(-\sum_c q_c \log q_c\right), \\ \mathrm{TV}(q, \bar{q}) = \tfrac{1}{2}\sum_c |q_c - \bar{q}_c|. \tag{10}$$

Lower ρ and TV and higher C_{eff} indicate weaker prediction collapse.

Table 8 shows that **LAMP-Merge consistently improves all collapse diagnostics**.

Table 6: DPR internal ablation by dataset. Each entry is reported as ACC / F1, where F1 denotes Macro-F1; all values average all backbones, client counts, and Dirichlet settings.

Category	Setting	Blood	Derma	Organ-C	Organ-S	US	Avg
Prototype semantics	CHA	18.95 / 7.94	56.14 / 12.15	17.74 / 8.78	18.88 / 8.47	16.43 / 9.36	25.63 / 9.34
	GFM	8.88 / 1.66	66.88 / 11.45	22.33 / 3.32	23.54 / 3.46	10.87 / 2.45	26.50 / 4.47
	SSH	11.62 / 5.87	17.29 / 7.57	7.08 / 4.55	7.74 / 4.11	10.69 / 6.77	10.89 / 5.77
	SLP	24.46 / 20.36	28.73 / 11.39	14.18 / 10.77	16.25 / 9.64	14.25 / 13.33	19.57 / 13.10
Client-support weighting	UCW	78.03 / 75.93	59.61 / 26.12	58.36 / 53.97	53.86 / 47.17	41.79 / 38.91	58.33 / 48.42
	BSO	78.03 / 75.93	44.96 / 27.03	57.86 / 54.69	53.24 / 48.56	41.79 / 38.91	55.18 / 49.02
	GCSW	78.45 / 76.13	62.48 / 25.84	59.11 / 54.55	53.79 / 46.79	43.13 / 40.38	59.39 / 48.74
	LAMP	**81.81 / 80.22**	**62.40 / 28.25**	**63.05 / 59.71**	**57.67 / 52.28**	**46.11 / 43.60**	**62.21 / 52.81**

Table 7: LPC internal ablation by dataset. UPP uses a uniform prior, AOC removes the activation gate, and CBP averages client-normalized priors. Each entry is reported as ACC / F1, where F1 denotes Macro-F1.

Setting	Blood	Derma	Organ-C	Organ-S	US	Avg
UPP	81.81 / 80.22	46.58 / 29.33	62.57 / 60.35	57.50 / 54.10	46.11 / 43.60	58.91 / 53.52
AOC	80.60 / 77.78	62.40 / 28.23	63.05 / 59.71	57.67 / 52.31	45.80 / 41.76	61.90 / 51.96
CBP	81.36 / 79.41	49.61 / 28.90	62.10 / 59.32	57.12 / 52.75	44.76 / 41.60	58.99 / 52.40
LAMP-Merge	**81.81 / 80.22**	**62.40 / 28.25**	**63.05 / 59.71**	**57.67 / 52.28**	**46.11 / 43.60**	**62.21 / 52.81**

Table 8: Prediction-collapse diagnostics by dataset. Best ref. is selected separately for each metric and dataset. Ratio metrics use percentage scale; Avg averages the five datasets.

Metric	Series	Blood	Derma	Organ-C	Organ-S	US	Avg
Collapse ↓	Best ref.	80.20	85.18	72.38	71.12	80.31	79.02
	LAMP	**19.58**	**68.05**	**20.72**	**24.36**	**20.45**	**30.63**
Eff. classes ↑	Best ref.	1.81	1.65	2.49	2.53	1.98	2.05
	LAMP	**7.48**	**3.10**	**9.55**	**8.44**	**7.44**	**7.20**
Pred-true TV ↓	Best ref.	73.48	46.03	75.42	73.85	72.73	69.44
	LAMP	**3.97**	**17.09**	**12.38**	**13.17**	**15.51**	**12.42**

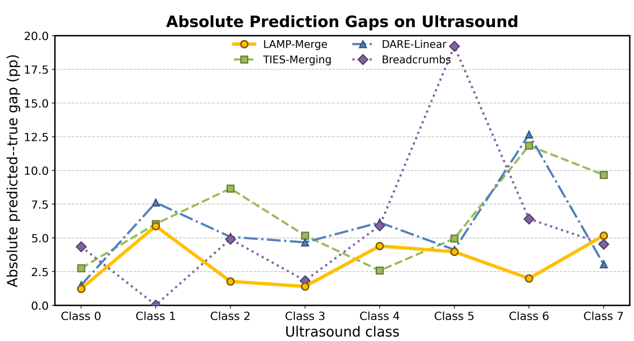


Figure 5: Absolute predicted–true class gaps on Ultrasound, averaged over all backbones, client counts, and Dirichlet settings. Values are percentage points; zero denotes exact agreement.

Visualization Experiment

We visualize the absolute per-class predicted–true gap; values closer to zero indicate better prevalence alignment.

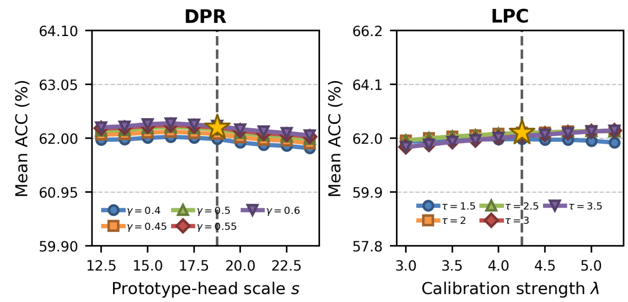


Figure 6: Full-grid DPR/LPC hyperparameter analysis on **the full setting**; each curve fixes one module parameter and scans the other over all datasets, backbones, client counts, and Dirichlet settings.

Hyperparameter Analysis Experiment

Fig. 6 evaluates DPR/LPC sensitivity on **the full setting**.

This full-grid analysis varies DPR's prototype-head scale s under several γ values and LPC's calibration strength λ under several τ values. The selected $s = 18.75$ remains inside a high-performing DPR region, and the LPC curves show gradual sensitivity rather than an isolated peak.

Conclusion

We study one-shot medical model merging without raw-image sharing or federated rounds. Under incomplete local coverage, generic merges can collapse predictions, while raw ACC may conceal failure on long-tailed data. Across full-scope experiments, DPR restores class-level directions and LPC retains clinically meaningful prevalence without reintroducing collapse. Together, they offer a class-level interface for privacy-constrained collaboration.

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